

Physics Mock Test

Atoms

Class 12 | UNIT 8 (A)

Marks: 24

Time: 60 Minutes

Coverage: Alpha-particle scattering, Rutherford's nuclear model, Bohr's atomic model, energy levels, and hydrogen spectrum series.

Instructions: For Section 2, use $R = 1.097 \times 10^7 \text{ m}^{-1}$ and $\pi \approx 3.14$. Show clear steps for energy conversions.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Rutherford's Nuclear Discovery

The alpha-particle scattering experiment provided the first evidence that:

- a) Atoms consist of negative nuclei
- b) Mass and positive charge are concentrated in a tiny nucleus
- c) Electrons are embedded in a positive sphere
- d) Atoms are solid, indivisible spheres

2. Bohr's Postulates

According to Bohr's model of the hydrogen atom, an electron can revolve only in those orbits where its angular momentum (L) is:

- a) Any integer multiple of $h/2\pi$
- b) An odd multiple of $h/4\pi$
- c) Equal to h/λ
- d) Independent of n

3. Energy Scale

The ground state energy of a hydrogen atom is -13.6 eV. What is the energy of the electron in the second excited state ($n=3$)?

- a) -3.4 eV
- b) -6.8 eV
- c) -1.51 eV
- d) -4.53 eV

4. Spectral Series Location

Which series of the hydrogen spectrum lies purely in the Ultraviolet (UV) region of the electromagnetic spectrum?

- | | |
|-------------------|-------------------|
| a) Lyman Series | b) Balmer Series |
| c) Paschen Series | d) Bracket Series |

5. Impact Parameter

In Rutherford scattering, if the impact parameter (b) is zero, the scattering angle (θ) will be:

- | | |
|----------------|---------------|
| a) 0° | b) 90° |
| c) 180° | d) 45° |

6. Spectral Line Transitions

A hydrogen atom in its third excited state ($n=4$) transitions to the ground state. The total number of spectral lines possible is:

- | | |
|------|-------|
| a) 3 | b) 4 |
| c) 6 | d) 10 |

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Calculation of Bohr Radius

Calculate the radius of the first Bohr orbit ($n=1$) of a hydrogen atom. Use the given constants to find the value in Angstroms or picometers.

8. Excitation Energy

Determine the energy required (in eV) to excite an electron in a hydrogen atom from the ground state ($n=1$) to the first excited state ($n=2$).

9. Lyman Series Wavelength

Calculate the wavelength of the longest wavelength line (limit line) in the Lyman series for hydrogen. Give your answer in nanometers (nm).

10. Quantization of Momentum

Find the orbital angular momentum of an electron in the third orbit ($n=3$) of a hydrogen atom according to Bohr's quantization condition.

11. Ionization Potential

Define ionization energy. Calculate the energy required to remove an electron completely from a hydrogen atom that is already in the second orbit ($n=2$).

12. Spectral Analysis

The wavelength of a spectral line is given by: $1/\lambda = R (1/2^2 - 1/n^2)$. Identify the name of this spectral series and calculate the wavelength for the transition $n = 3$.

Answer Key & Solutions

1. b) Mass and charge concentrate in nucleus

As shown by large-angle scattering of alpha particles.

2. a) Any integer multiple of $h/2\pi$

$$mvr = nh/2\pi.$$

3. c) -1.51 eV

$$E = -13.6 / 3^2 = -13.6 / 9 = -1.51 \text{ eV}.$$

4. a) Lyman Series

Transitions to $n=1$ result in high energy UV photons.

5. c) 180°

Head-on collision results in complete reversal of direction.

6. c) 6

$$\text{Number of lines} = n(n-1)/2 = 4(3)/2 = 6.$$

7. $\approx 0.529 \text{ \AA}$ (52.9 pm)

$$r_n = \epsilon_0 h^2 / (\pi m e^2) \approx 0.529 \text{ \AA}.$$

8. 10.2 eV

$$\Delta E = E_2 - E_1 = (-3.4) - (-13.6) = 10.2 \text{ eV}.$$

9. $\approx 121.5 \text{ nm}$

$$1/\lambda = R(1/1^2 - 1/2^2) = R(3/4). \lambda = 4 / (3 \times 1.097 \times 10^7) \approx 1.215 \times 10^{-7} \text{ m} = 121.5 \text{ nm}.$$

10. $3.16 \times 10^{-34} \text{ J}\cdot\text{s}$

$$L = 3h / 2\pi = (3 \times 6.63 \times 10^{-34}) / 6.28 \approx 3.16 \times 10^{-34} \text{ J}\cdot\text{s}.$$

11. 3.4 eV

E at $n=2$ is -3.4 eV. Energy required to reach 0 eV (infinity) is 3.4 eV.

12. Balmer Series | 656.3 nm

$$1/\lambda = R(1/4 - 1/9) = R(5/36). \lambda = 36 / (5R) \approx 656.3 \text{ nm}.$$